



General Characteristics

221. What is the name of element with atomic number 105
 (a) Kurchatovium (b) Dubnium
 (c) Nobelium (d) Holmium
222. Electrons in a paramagnetic compound are
 (a) Shared (b) Unpaired
 (c) Donated (d) Paired
223. Which of the following pairs involves isoelectronic ions
 (a) Mn^{3+} and Fe^{2+}
 (b) Mn^{2+} and Fe^{3+}
 (c) Cr^{3+} and Fe^{2+}
 (d) Fe^{2+} and Co^{2+}
224. Which of the following is paramagnetic
 (a) Ni^{++} (b) Cu^{+}
 (c) Zn^{++} (d) Sc^{+++}
225. The electronic configuration of chromium is
 (a) $[Ne]3s^23p^63d^4, 4s^2$
 (b) $[Ne]3s^23p^63d^5, 4s^1$
 (c) $[Ne]3s^23p^63d^6, 4s^1$
 (d) $[Ne]3s^23p^53d^5, 4s^2$
226. Electronic configuration of Cu ($Z = 29$) is
 (a) $[Ar]3d^94s^2$
 (b) $[Ar]3d^{10}4s^1$
 (c) $[Ar]3d^54s^2$
 (d) $[Ar]3d^64s^2$
227. Ce-58 is a member of
 (a) s-block (b) p-block
 (c) d-block (d) f-block
228. How many unpaired electrons are there in
 (a) 2 (b) 4
 (c) 5 (d) 0
229. The main reason for larger number of oxidation states exhibited by the actinoids than the corresponding lanthanoids is
 (a) Lesser energy difference between 5f and 6d orbitals than between 4f and 5d orbitals
 (b) Larger atomic size of actinoids than the lanthanoids
 (c) More energy difference between 5f and 6d orbitals than between 4f and 5d orbitals
 (d) Greater reactive nature of the actinoids than the lanthanoids
230. Four successive members of the first row transition elements are listed below with their atomic numbers. Which one of them is expected to have the highest third ionization enthalpy
 (a) Vanadium ($Z = 23$)
 (b) Chromium ($Z = 24$)
 (c) Iron ($Z = 26$)
 (d) Manganese ($Z = 25$)



231. The aqueous solution containing which one of the following ions will be colourless
(a) Sc^{3+} (b) Fe^{2+}
(c) Ti^{3+} (d) Mn^{2+}
(Atomic number $Sc = 21$, $Fe = 26$, $Ti = 22$, $Mn = 25$)
232. Which of the following trivalent ion has the largest atomic radii in the lanthanide series
(a) La (b) Ce
(c) Pm (d) Lu
233. Which of the following does not have valence electron in $3d$ -subshell
(a) Fe (III) (b) Mn (II)
(c) Cr (I) (d) P (0)
234. Among the following pairs of ions, the lower oxidation state in aqueous solution is more stable than the other in
(a) Tl^+ , Tl^{3+} (b) Cu^+ , Cu^{2+}
(c) Cr^{2+} , Cr^{3+} (d) V^{2+} , VO^{2+}
235. The lanthanide contraction is responsible for the fact that
(a) Zr and Y have about the same radius
(b) Zr and Nb have similar oxidation state
(c) Zr and Hf have about the same radius
(d) Zr and Zn have the same oxidation state
236. Which of the following factors may be regarded as the main cause of lanthanide contraction
(a) Poor shielding of one of $4f$ electron by another in the subshell
(b) Effective shielding of one of $4f$ electrons by another in the subshell
(c) Poorer shielding of $5d$ electrons by $4f$ electrons
(d) Greater shielding of $5d$ electron by $4f$ electrons
237. Which of the following have maximum number of unpaired electrons
(a) Fe^{3+} (b) Fe^{2+}
(c) Co^{2+} (d) Co^{3+}
238. Transition metals show paramagnetism
(a) Due to characteristic configuration
(b) High lattice energy
(c) Due to variable oxidation states
(d) Due to unpaired electrons
239. Which of the following pairs of elements cannot form an alloy
(a) Zn, Cu (b) Fe, Hg
(c) Fe, C (d) Hg, Na
240. Which belongs to the actinides series
(a) Ce (b) Cf
(c) Ca (d) Cs



241. Effective magnetic moment of Sc^{+3} ion is

- (a) 1.73 (b) 0
(c) 5.92 (d) 2.83
(e) 3.87

